



Recurring Billing Module ERD

Entity relationships for BC Subscription Billing
from microsoft/BCApps source

VERSION	1.0
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CLASSIFICATION	Internal
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The Recurring Billing module (officially "Subscription Billing") was added to BC as a first-party app. It handles subscription contracts, recurring invoicing, revenue deferrals, price updates, and contract renewals. This document maps the entity structure.

1. Entity overview — 44 tables in 7 layers

LAYER	PURPOSE	KEY TABLES
Setup	Configuration and templates	Subscription Contract Setup, Subscription Contract Type, Sub. Package Line Template
Package	Reusable subscription definitions	Subscription Package, Sub. Package Line, Item Subscription Package
Subscription	The service object + monetary lines	Subscription Header, Subscription Line
Contract	Customer/vendor agreements	Customer/Vendor Subscription Contract, Contract Lines
Billing	Invoice generation	Billing Template, Billing Line, Billing Line Archive
Deferral	Revenue/cost recognition	Cust./Vend. Contract Deferral
Lifecycle	Renewals, price updates, analysis	Planned Subscription Line, Price Update Template, Sub. Contract Analysis Entry

2. ERD — core entities and relationships



2.1 The key flow (how entities connect end-to-end)



2.2 Item Subscription Option (the starting point)

Items get a new field **Subscription Option** that controls their behavior:

VALUE	MEANING
No Subscription	Normal item — no subscription behavior
Sales with Subscription	Item sold normally + subscription lines auto-created from linked package
Subscription Item	Item exists only as a subscription — no standalone sale

VALUE	MEANING
Invoicing Item	Used as the invoice line item when billing subscription contract lines

2.3 Subscription Package → Subscription Line flow

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Subscription Package (template)
├── Sub. Package Lines (1:N)
│   ├── Partner: Customer or Vendor
│   ├── Invoicing Via: Sales or Subscription Contract
│   ├── Calculation Base: Item Price / Document Price / Doc Price + Discount
│   ├── Billing Base Period: e.g., 1M
│   ├── Billing Rhythm: e.g., 1M
│   ├── Initial Term: e.g., 1Y
│   ├── Extension Term: e.g., 1Y (auto-renewal)
│   └── Notice Period: e.g., 3M
│
│   ↓ (instantiated on shipment)

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Subscription Line (on a Subscription Header)
├── Same fields as above, now with actual dates
├── Next Billing Date
├── Calculation Base Amount → Price → Amount
└── Assigned to a Contract Line

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2.4 Contract structure

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Customer Subscription Contract
├── Sell-to Customer No.
├── Bill-to Customer No.
├── Contract Type (→ Sub. Contract Type: harmonized billing, deferrals)
├── Currency, Payment Terms, Salesperson
├── Billing Base Date (for harmonized billing across lines)
├── Active (Boolean)
└── Cust. Sub. Contract Lines (1:N)
    ├── Contract Line Type: Comment | Item | G/L Account
    ├── → Subscription Header (the subscription)
    └── → Subscription Line (the monetary commitment)

```

2.5 Billing flow

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Billing Template
├── Partner: Customer or Vendor
├── Billing Date/To Date Formulas
├── Group by: None | Contract | Contract Partner
├── Customer Document per: Contract | Sell-to | Bill-to
├── Automation: None | Create Billing Proposal and Documents
└── generates → Billing Lines (1:N)
    └── For each contract line with due billing

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- Billing From / To dates
- Amount
- → Document Type: Invoice or Credit Memo
- → Document No. (Sales/Purchase Header)
- → Document Line No. (Sales/Purchase Line)

Billing Template automation can run on a schedule (Minutes between runs) to auto-generate proposals and documents.

2.6 Revenue deferrals

When **Create Contract Deferrals = Yes** on the contract type (or overridden per line):

- Posting the invoice creates **Cust. Contract Deferral** entries
- Each entry represents a portion of revenue for a specific period
- Deferrals are released on their posting date → creates **G/L Entries** spreading revenue across the contract period
- Same pattern on vendor side for cost deferrals

2.7 Contract lifecycle

Renewals: Before a contract line's term expires, a **Planned Subscription Line** is created to stage the renewal. When the current line is fully invoiced, the planned line replaces it.

Price updates: **Price Update Template** defines the method (% of calculation base, % of price, or recent item prices). Generates **Price Update Lines** as a worksheet for review before applying.

Analysis: **Sub. Contract Analysis Entry** captures snapshots for contract analytics, tagged by type (None, Contract Renewal, Price Update).

3. Connections to core BC

CORE BC ENTITY	HOW THE MODULE CONNECTS
Customer	Sell-to / Bill-to on contracts and subscriptions
Vendor	Buy-from / Pay-to on vendor contracts
Item	Source of subscription (Subscription Option field), invoicing item on lines
G/L Account	Alternative source type on subscriptions; line type on contracts
Sales Header/Line	Billing Lines generate sales invoices/credit memos
Purchase Header/Line	Billing Lines generate purchase invoices/credit memos

CORE BC ENTITY	HOW THE MODULE CONNECTS
G/L Entry	Deferrals post to G/L on release
Customer Price Group	Drives pricing on packages, subscriptions, contracts
Contact	Linked to contracts via Sell-to/Bill-to Contact No.
Dimension Set Entry	Contracts and subscription lines carry dimension sets
Job	Contracts can inherit dimensions from a Job

4. Usage-based billing extension

The module includes a **usage-based billing** sub-system for metered/consumption pricing:

TABLE	PURPOSE
Usage Data Supplier	External service providers (e.g., Azure, AWS)
Usage Data Import	Import headers for usage data
Usage Data Billing	Billing data generated from usage records
Usage Data Supplier Reference	Maps supplier IDs to BC subscriptions
Usage Data Generic Import	Generic CSV/file import format

This extends the flat-rate subscription model to support per-unit, tiered, or consumption-based pricing.

Version History

VERSION	DATE	CHANGES
1.0	2026-06-08	Initial ERD from microsoft/BCApps source